

Household state | Question | Options A / B / C → **Six parallel decision interfaces**

15 models

Encoder option scoring

Encoder → p(a)

Base Laya / YN-mini
Laya-Finance / Von 1.3
OpenDecider Nano

LM + learned decision head

Decoder LM → Head

Kev 4B
Clef Flash 9B

Next-token decision SFT

Decoder LM → Token

Tev1-4B
Existing LM output head

Dynamic schema classification

{labels} → Classifier

GLiNER2.5-Decide
Caller-supplied labels

Multimodal encoder decisions

Encoder → Head

Liquid d1-omni-600M
Text input in this test

Provider decision interfaces

Jev 1.13: structured API

GLM-5.3-Flash
DeepSeek-V4.1-Flash
GPT-6-Luna / Astra

Generated outputs → fixed options

Shared full test: 240,458 records, 119 question versions; accuracy and question-macro F1.
14 interface models and all 10 variants + Original Laya (no new training). Astra: paired 500 groups / 14,441 records.

Five encoder training variants

real-survey data → training QA

Original Laya

- 01 **SFT 50k** unique examples
  - 02 **SFT → RLCD** +50k pass
- 03 **SFT 200k** unique examples
  - 04 **SFT → RLCD** +200k pass
  - 05 **SFT → CE** +200k pass

RLCD = CE + proper-score policy gradient on noisy option logits.
Reward: target-answer score.

Each run selects its checkpoint on a fixed 20k development set.

Full-data reference 2.93M SFT+RLCD — **YN-mini** — MLX execution format → **YN-mini-MLX**

Five decoder training variants

Qwen3.5-4B-Base

- **Head-only** Train head; freeze backbone
  - 06 **50k**
  - 07 **200k**
- **LoRA + head** Train adapters + scoring head
  - 08 **50k**
  - 09 **200k** — YN-Lite-4B

412,873 full-dev records compare both 4B 200k methods; LoRA + head selected.

Qwen3.5-9B Post-trained Independent init.

- 10 **LoRA + head** 200k — YN-master-9B

Method choice, not weights

50k / 200k = unique examples; +50k / +200k = one more pass over the same examples.

Each candidate encodes the full question and household state

Candidate encoding → L2 → Shared scalar head → Option softmax